Section 3.4: Solving System of Linear Equations by using Gauss-Jordan Elimination (Reference: Sections 2.1, 2.2)

1) <u>A linear equation in the *n* variables</u> (or unknowns) $x_1, x_2, \cdots, x_n$ is an equation that can be expressed in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \cdots, a_n$ and b are real constants.
<u>A solution of this linear equation</u> is a sequence of *n* numbers $s_1, s_2, \cdots, s_n$ such that the equation is satisfied when we substitute $x_1 = s_1, x_2 = s_2, \cdots, x_n = s_n$.
For example, the linear equation $2x_1 - 3x_2 + 4x_3 = 1$ has a solution $x_1 = 1, x_2 = 3, x_3 = 2$ since $2(1) - 3(3) + 4(2) = 1$.
The set of all solutions of the equation is called <u>its solution set</u> or sometimes called <u>the general solution</u> of the equation.

2) A finite set of *m* linear equations in the variables $x_1, x_2, \cdots, x_n$ is called <u>a system of linear equations</u> or <u>a linear system</u>:

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \cdots & + & a_{nn}x_n & = & b_n \end{array}$$

where the subscripted *a*'s and *b*'s denote constants. For the double subscripting on the coefficients of the variables, the first subscript on the coefficient a_{ij} indicates the equation in which the coefficient occurs, and the second subscript indicates which variable it multiplies. A sequence of numbers $s_1, s_2, \cdots, s_n$ is called <u>a solution of the system</u> if $x_1 = s_1, x_2 = s_2, \cdots, x_n = s_n$ is a solution of every equation in the system. For example, the linear system $\left\{\begin{array}{ccccccc} x_1 & + & 2x_2 & + & 3x_3 & = & 14 \\ 3x_1 & + & 4x_2 & + & 5x_3 & = & 26 \end{array}\right\}$ has a solution $x_1 = 3, x_2 = -2. x_3 = 5$, since $\left\{\begin{array}{ccccccc} (3) & + & 2(-2) & + & 3(5) & = & 14 \\ 3(3) & + & 4(-2) & + & 5(5) & = & 26 \end{array}\right\}$.
The collection of solutions of the linear system is <u>its solution set</u>.
Two linear systems are <u>equivalent</u> if they have the same solution set.

3) This linear system can be abbreviated by writing only the rectangular array of numbers:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{bmatrix} \quad \text{or} \quad \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right].$$

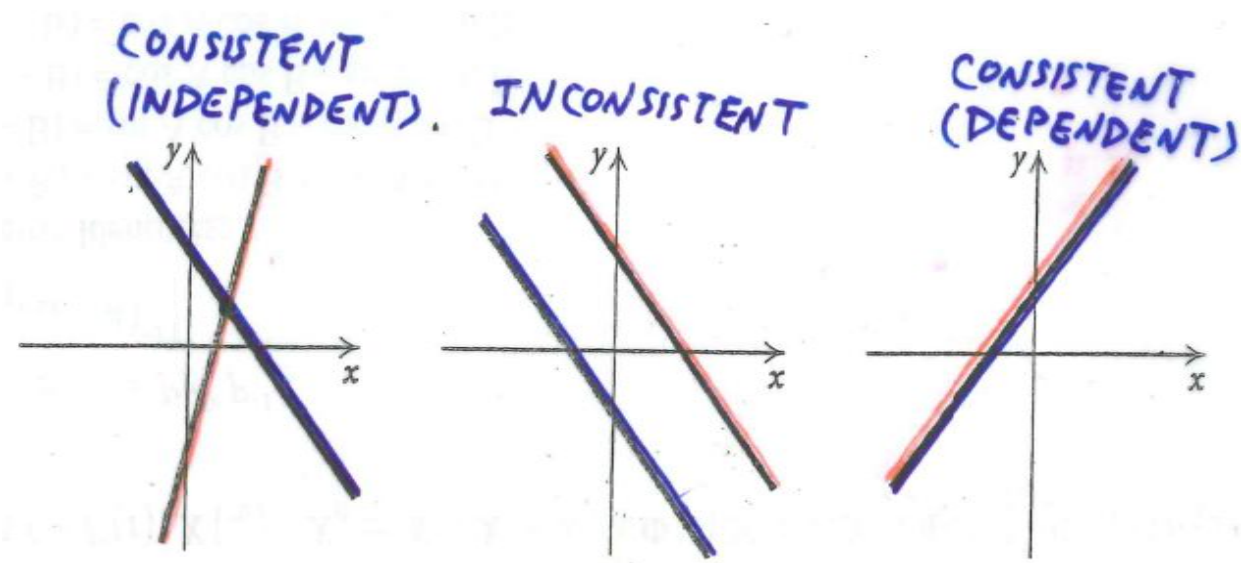
This is called the <u>augmented matrix</u> for the system. Note that when constructing an augmented matrix, we must write the variables in the same order in each equation, and the constants must be on the right.
We shall discuss matrix in more details in later sections.

4) A matrix is in <u>reduced row-echelon form</u> (rref) if it has the following 4 properties:
(i) If a row does not consist entirely of zeros, then the first nonzero number in the row is a 1. We call this <u>a leading one</u>.
(ii) If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
(iii) In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.
(iv) Each column that contains a leading 1 has zeros everywhere else in that column.
Remarks:

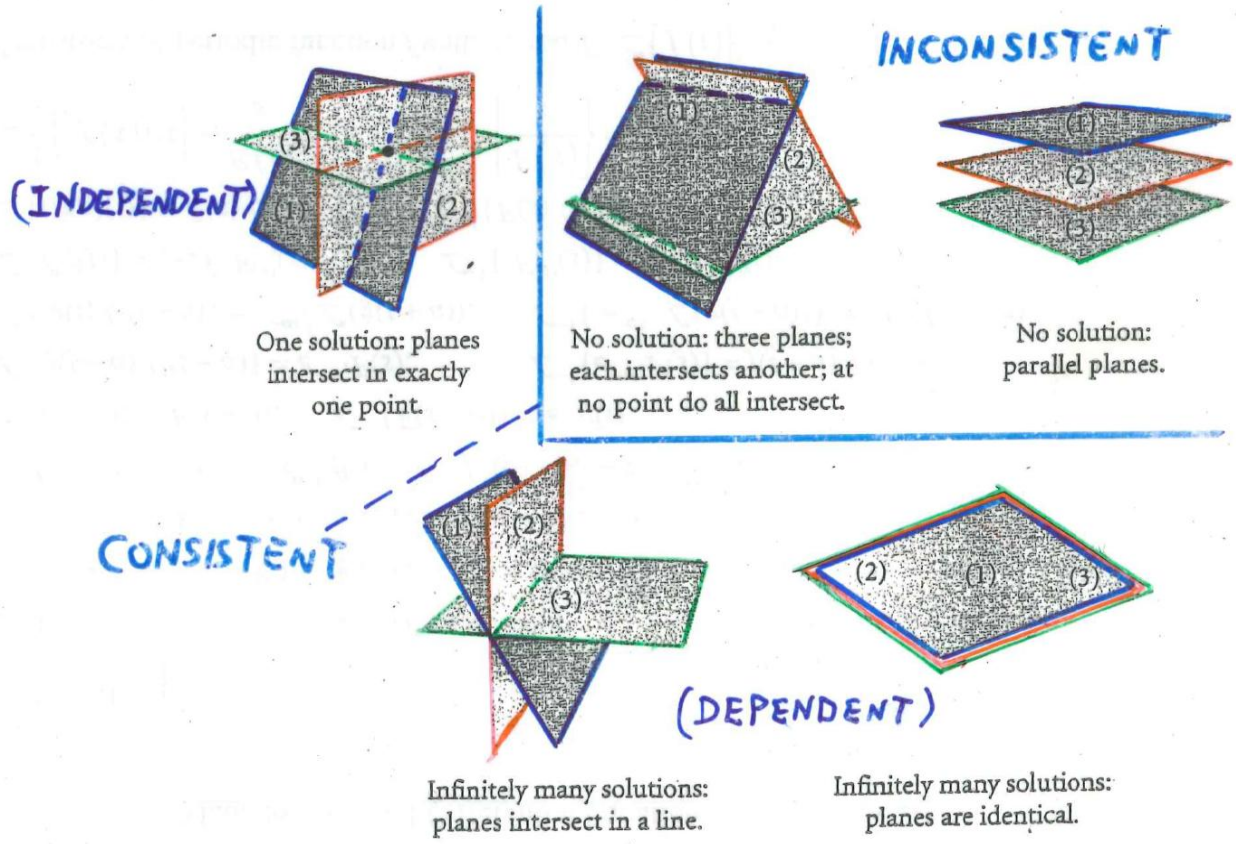
- A matrix in reduced row-echelon form has zeros below and above each leading 1.
- Every matrix has a unique reduced row-echelon form.

5) Every system of linear equations has no solution, or has exactly one solution, or has infinitely many solutions (for example, consider the number of solutions of a linear system of 2 equations in 2 variables). A linear system that has no solutions is said to be <u>inconsistent</u>; if there is at least one solution of the system, it is called <u>consistent</u>. The equations of a consistent system are <u>independent</u> (or <u>dependent</u>, respectively) if there is a unique solution (or an unlimited number of solutions, respectively).

In particular, when we solve a system of two equations in two variables, the solution(s) of the system corresponds to the points of intersection of the two lines represented by the two equations, and we can have one intersection point, no intersection point, or infinitely many intersection points:



When we solve a system of three equations in three variables, the solution(s) of the system corresponds to the points of intersection of the three planes represented by the three equations, and we can have one intersection point, no intersection point, or infinitely many intersection points:



6) When a linear equation has more than one solution, we describe its general solution in terms of one or more arbitrary number called <u>parameter</u>. Particular solutions can be obtained by substituting specific values for the parameter(s).

7) For a system of linear equations with coefficient matrix A, the <u>free variables</u> correspond to the columns that do not contain leading 1 in rref(A), the variables that correspond to the leading 1's are called <u>leading variables</u> or (<u>pivots</u>),

8) To solve a linear system by using the reduced row-echelon form of its augmented matrix:
Step 1: Assign arbitrary parameter values to the free variables, if any.
Step 2: Solve the leading variables, possibly in terms of the parameter values.

9) The valid operations we perform on the augmented matrix of a linear system to get the reduced row-echelon form are called the <u>elementary operations</u>. There are three types, each corresponds to an operation to create an equivalent linear system:

Operation on Linear System	Row Operation on Augmented Matrix
1. Multiply the ith equation through by a nonzero constant k.	1. Multiply row i through by k, . denoted by kR_i.
2. Interchange the ith and the jth equations.	2. Interchange row i and row j, denoted by $R_i \leftrightarrow R_j$.
3. Add k times the ith equation to the jth equation.	3. Add k times row i to row j, denoted by $R_j + kR_i$.

Note: These are **<u>not</u>** valid row operation (as the new matrix they create correspond to a linear system with different solution):

- Add a constant to each entry of a row
- Multiply row i to row j

10) Converting an augmented matrix to its reduced row-echelon form.

Step 1. Locate the leftmost column that does not consist entirely of zeros.

Step 2. Interchange the top row with another row, if necessary, to bring a nonzero entry to the top of the column found in step 1.

Step 3. If the entry that is now at the top of the column found in step 1 is a, multiply the first row by $1/a$ in order to introduce a leading 1.

Step 4. Add suitable multiples of the top row to the rows below so that all entries below the leading 1 become zeros.

Step 5. Now cover the top row in the matrix and begin again with step 1 applied to the submatrix that remains. Continue in this way until we reach the bottom of the entire matrix.

Step 6. beginning with the last nonzero row and working upward, add suitable multiples of each row to the rows above to introduce zero above the leading 1's.

11) The procedure that produces a matrix in reduced row-echelon form is called **Gauss-Jordan elimination**.

Example 3.4.1. Use Gauss-Jordan Elimination to solve the system of linear equations

$$\begin{Bmatrix} x & + & 2y & + & 3z & = & 14 \\ 2x & + & 3y & + & 4z & = & 20 \\ 3x & + & 4y & + & 5z & = & 26 \end{Bmatrix}$$

Solution: (Set up augmented matrix for the linear system and use elementary row operations to get its RREF)

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 20 \\ 3 & 4 & 5 & 26 \end{array}\right] \xrightarrow[\square]{R_2+(-2)R_1, R_3+(-3)R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -2 & -8 \\ 0 & -2 & -4 & -16 \end{array}\right] \rightarrow$$

$$\xrightarrow[\square]{(-1)R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & -2 & -4 & -16 \end{array}\right] \xrightarrow[\square]{R_3+(2)R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[\square]{R_1+(-2)R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

(Solve for the leading variables, possibly in terms of the free variable(s))

$$\begin{Bmatrix} x - z = -2 \Longrightarrow x = z - 2 \\ y + 2z = 8 \Longrightarrow y = 8 - 2z \\ \text{Free Variable } z = t \end{Bmatrix} \Longrightarrow \boxed{\begin{Bmatrix} \boldsymbol{x = t - 2} \\ \boldsymbol{y = 8 - 2t} \\ \boldsymbol{z = t} \end{Bmatrix}}$$

Note: There are multiple ways to write our solution, for instance, we can use the parameter t to represent any of the variables, hence $\begin{Bmatrix} x = t - 2 \\ y = 8 - 2t \\ z = t \end{Bmatrix}$, $\begin{Bmatrix} x = 2 - \frac{1}{2}t \\ y = t \\ z = 4 - \frac{1}{2}t \end{Bmatrix}$, and $\begin{Bmatrix} x = t \\ y = 4 - 2t \\ z = t + 2 \end{Bmatrix}$ are all equivalent solutions. Also, in the second solution, we can reduce the number of fractions involved by choosing $y = 2t$: $\begin{Bmatrix} x = 2 - t \\ y = 2t \\ z = 4 - t \end{Bmatrix}$ is hence another solution.

Example 3.4.2. Use Gauss-Jordan Elimination to solve the system of linear equations.

$$\begin{Bmatrix} (i)x & + & y & + & 2z & = & 2 - i \\ (-3i)x & - & 3y & - & 6z & = & -6 + 3i \\ -2x & + & (2i)y & + & (4i)z & = & 2 + 4i \end{Bmatrix}$$

Solution: $\left[\begin{array}{ccc|c} i & 1 & 2 & 2 - i \\ -3i & -3 & -6 & -6 + 3i \\ -2 & 2i & 4i & 2 + 4i \end{array}\right]$

[Since division of i is easy – multiply both numerator and denominator by its conjugate $-i$, we can introduce our first leading one by dividing the first row by i. Another option would be interchanging row 1 and row 3 and then divide row 1 by -2.]

$$\xrightarrow[\square]{\left(\frac{1}{i}\right)R_1} \left[\begin{array}{ccc|c} 1 & -i & -2i & -1 - 2i \\ -3i & -3 & -6 & -6 + 3i \\ -2 & 2i & 4i & 2 + 4i \end{array}\right] \xrightarrow[\square]{R_2+(3i)R_1, R_3+(2)R_1} \left[\begin{array}{ccc|c} 1 & -i & -2i & -1 - 2i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{Bmatrix} x - iy - 2iz = -1 - 2i \Longrightarrow x = -1 - 2i + iy + 2iz \\ \text{Free Variable } y = s \\ \text{Free Variable } z = t \end{Bmatrix} \Longrightarrow \boxed{\begin{matrix} \boldsymbol{x = -1 - 2i + is + 2it} \\ \boldsymbol{y = s} \\ \boldsymbol{z = t} \end{matrix}}$$

12) A vector $\vec{\mathbf{v}}$ in $\mathbb{R}^n$ is a <u>linear combination</u> of vectors $\vec{\mathbf{v}}_1, \ldots, \vec{\mathbf{v}}_m$ in $\mathbb{R}^n$ if there exists scalars $k_1, \cdots, k_m$ such that

$$\vec{\mathbf{v}} = k_1\vec{\mathbf{v}}_1 + \cdots + k_m\vec{\mathbf{v}}_m.$$

The scalars $k_1, \cdots, k_m$ are called <u>the coordinates of $\vec{\mathbf{v}}$ with respect to $\vec{\mathbf{v}}_1, \cdots, \vec{\mathbf{v}}_m$</u>.

Example 3.4.3. Find the coordinates of $\vec{\mathbf{v}} = [-9, \quad -11, \quad 8]$ with respect to $\vec{\mathbf{v}}_1 = [2, \quad 4, \quad -2]$,. $\vec{\mathbf{v}}_2 = [-1, \quad -1, \quad -1]$, and $\vec{\mathbf{v}}_3 = [3, \quad -1, \quad -2]$.

Solution: [By definition, we need to find scalars k_1, k_2, and k_3 so that

$\vec{\mathbf{v}} = k_1\vec{\mathbf{v}}_1 + k_2\vec{\mathbf{v}}_2 + k_3\vec{\mathbf{v}}_3$.

We perform the vector operations involved on the right hand side.]

$\vec{\mathbf{v}} = k_1\vec{\mathbf{v}}_1 + k_2\vec{\mathbf{v}}_2 + k_3\vec{\mathbf{v}}_3$

$\Rightarrow [-9, \quad -11, \quad 8] = k_1[2, \quad 4, \quad -2] + k_2[-1, \quad -1, \quad -1] + k_3[3, \quad -1, \quad -2]$

$\Rightarrow [-9, \quad -11, \quad 8] = [2k_1, \quad 4k_1, \quad -2k_1] + [-k_2, \quad -k_2, \quad -k_2] + [3k_3, \quad -k_3, \quad -2k_3]$

$\Rightarrow [-9, \quad -11, \quad 8] = [2k_1 - k_2 + 3k_3, \quad 4k_1 - k_2 - k_3, \quad -2k_1 - k_2 - 2k_3]$

[By comparing these two vectors one component at a time, then we have a system of 3 linear equations. We find the coordinates by solving this linear system.]

$$\left\{\begin{array}{rcrcrcr} 2k_1 & - & k_2 & + & 3k_3 & = & -9 \\ 4k_1 & - & k_2 & - & k_3 & = & -11 \\ -2k_1 & - & k_2 & - & 2k_3 & = & 8 \end{array}\right\}$$

$$\left[\begin{array}{rrr|r} 2 & -1 & 3 & -9 \\ 4 & -1 & -1 & -11 \\ -2 & -1 & -2 & 8 \end{array}\right] \xrightarrow[]{\left(\frac{1}{2}\right)R_1} \left[\begin{array}{rrr|r} 1 & -1/2 & 3/2 & -9/2 \\ 4 & -1 & -1 & -11 \\ -2 & -1 & -2 & 8 \end{array}\right]$$

$$\xrightarrow{R_2+(-4)R_1, R_3+(2)R_1} \left[\begin{array}{rrr|r} 1 & -1/2 & 3/2 & -9/2 \\ 0 & 1 & -7 & 7 \\ 0 & -2 & 1 & -1 \end{array}\right] \xrightarrow{R_3+(2)R_2} \left[\begin{array}{rrr|r} 1 & -1/2 & 3/2 & -9/2 \\ 0 & 1 & -7 & 7 \\ 0 & 0 & -13 & 13 \end{array}\right]$$

$$\xrightarrow{\left(-\frac{1}{13}\right)R_3} \left[\begin{array}{rrr|r} 1 & -1/2 & 3/2 & -9/2 \\ 0 & 1 & -7 & 7 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_1+\left(-\frac{3}{2}\right)R_3, R_2+(7)R_3} \left[\begin{array}{rrr|r} 1 & -1/2 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$$\xrightarrow{R_1+\left(\frac{1}{2}\right)R_2} \left[\begin{array}{rrr|r} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array}\right] \Rightarrow k_1 = -3, k_2 = 0, k_3 = -1$$

Hence $\vec{\mathbf{v}} = (-3)\vec{\mathbf{v}}_1 + 0\vec{\mathbf{v}}_2 + (-1)\vec{\mathbf{v}}_3$, and

$\boxed{\text{the coordinates of } \vec{\mathbf{v}} \text{ with respect to } \vec{\mathbf{v}}_1,. \vec{\mathbf{v}}_2, \text{ and } \vec{\mathbf{v}}_3 \text{ are } (-3, 0, -1).}$

Note: Changing the orders of the $\vec{\mathbf{v}}_i's$ changes the answer; for instance, since $\vec{\mathbf{v}} = (-1)\vec{\mathbf{v}}_3 + (-3)\vec{\mathbf{v}}_1 + 0\vec{\mathbf{v}}_2$, the coordinates of $\vec{\mathbf{v}}$ with respect to $\vec{\mathbf{v}}_3,. \vec{\mathbf{v}}_1$, and $\vec{\mathbf{v}}_2$ are $(-1, -3, 0)$.

Practice Exercises

1. Use Gauss-Jordan Elimination to solve the system of linear equations

(a) $\begin{Bmatrix} 3x & - & y & + & 2z & = & 3 \\ 4x & - & 2y & + & z & = & 3 \\ 6x & + & 6y & + & 3z & = & 4 \end{Bmatrix}$ |

(b) $\begin{Bmatrix} 2x & + & 4y & + & 6z & = & -5 \\ x & + & 3y & - & 4z & = & -5 \\ -x & + & y & - & 24z & = & -5 \end{Bmatrix}$

(c) $\begin{Bmatrix} x & + & y & \square & \square & = & -1 \\ -3x & - & 2y & + & 4z & = & 2 \\ x & \square & \square & - & 4z & = & 3 \end{Bmatrix}$

(d) $\begin{Bmatrix} \square & \square & 4y & + & 3z & = & -2+4i \\ 4x & + & 4y & - & z & = & -14+16i \\ x & + & 4y & + & (2i)z & = & -9+11i \end{Bmatrix}$

(e) $\begin{Bmatrix} (2+2i)x & + & 4y & + & 4z & = & 0 \\ -x & + & (-2+2i)y & \square & \square & = & 0 \\ -x & \square & \square & - & (2-2i)z & = & 0 \end{Bmatrix}$

(f) (Assignment 2) $\begin{Bmatrix} a & + & b & + & c & + & d & = & 4 \\ a & + & 2b & + & 3c & + & d & = & 14 \\ 2a & + & 2b & + & 3c & + & 3d & = & 9 \\ 7a & + & 8b & + & 10c & + & 8d & = & 39 \end{Bmatrix}$

[Answers: (a) $x = \frac{1}{2}, y = -\frac{1}{6}, z = \frac{2}{3}$

(b) There are many possible answers, such as $x = \frac{5}{2} - 17t, y = 7t - \frac{5}{2}, z = t$

(c) The system has no solution. (d) $x = -1 + 3i, y = -2 + i, z = 2$

(e) There are many possible answers, such as $x = (-2+2i)t, y = t, z = t$

(f) $a = -5 - 2t, b = 8 + 2t, c = 1 - t, d = t$]

2. Write $\vec{\mathbf{w}} = \begin{bmatrix} 8 \\ 21 \\ 15 \end{bmatrix}$ as a linear combination of $\vec{\mathbf{u}} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{\mathbf{v}} = \begin{bmatrix} 4 \\ -3 \\ 5 \end{bmatrix}$ or show that it is not possible to do so.

[Answer: It is not possible as $k_1\vec{\mathbf{u}} + k_2\vec{\mathbf{v}} = \vec{\mathbf{w}}$ has no solution.]

3. Find the coordinates of $\vec{\mathbf{v}}$ with respect to $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$.

(a) $\vec{\mathbf{v}} = [4, \quad -18], \vec{\mathbf{v}}_1 = [4, \quad -4], \vec{\mathbf{v}}_2 = [4, \quad 1]$

(b) $\vec{\mathbf{v}} = [5, \quad -4, \quad -3], \vec{\mathbf{v}}_1 = [-3, \quad 2, \quad 2], \vec{\mathbf{v}}_2 = [-4, \quad 2, \quad 3]$

[Answers: (a) $(4, -3)$ (b) $(-3, 1)$]

4. Find the coordinates of $\vec{\mathbf{v}} = [-25, \quad -6, \quad 2]$ with respect to $\vec{\mathbf{v}}_1 = [-4, \quad -2, \quad -3]$,. $\vec{\mathbf{v}}_2 = [-3, \quad -2, \quad 1]$, and $\vec{\mathbf{v}}_3 = [-1, \quad 2, \quad 4]$.

[Answer: $(4, 2, 3)$]